

## ***Project 1 : Spotify – YouTube Combined Dataset***

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***Course / Subject: Power BI***

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# 1. INTRODUCTION

Data cleaning is a critical step in ensuring a dataset is usable, consistent, accurate, and ready for meaningful analysis.

This project focuses on cleaning a combined **Spotify–YouTube dataset** using **Microsoft Power BI**, specifically the **Power Query Editor**.

The original dataset contained many issues, including:

- Missing values
- Blank values
- Incorrect data types
- Merged/combined columns
- Duplicate rows
- Random columns
- Inconsistent formatting
- Invalid text entries in numeric columns

The goal of this project is to systematically clean the dataset and prepare it for further analysis and visualization.

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# 2. OBJECTIVE OF THE PROJECT

The major objectives are:

- Clean and prepare the raw Spotify–YouTube dataset
- Identify and fix missing values

- Standardize all column names
  - Split merged columns
  - Convert data types to appropriate formats
  - Remove invalid or irrelevant data
  - Ensure consistency for analysis
  - Make the dataset suitable for a Power BI dashboard
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### 3. DATASET OVERVIEW

The dataset includes:

- **Spotify audio features**  
(danceability, energy, valence, tempo, acousticness, etc.)
- **YouTube performance data**  
(views, likes, comments, channel data, official video flag)
- **Track metadata**  
(artist, track name, album, album type)

The dataset originally contained **multiple data quality issues**, which required intensive cleaning before use.

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### 4. TOOLS USED

| Tool                     | Purpose                           |
|--------------------------|-----------------------------------|
| Power BI Desktop         | Main platform for cleaning        |
| Power Query Editor       | Data transformation               |
| Power BI Query Functions | Trim, Clean, Replace, Split, etc. |
| Column Quality Tools     | Detect errors, blanks, nulls      |

| Tool                | Purpose             |
|---------------------|---------------------|
| Data Type Converter | Changing data types |

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## 5. IMPORTING THE DATASET

Steps:

1. Open **Power BI Desktop**
2. Go to **Home → Get Data → Text/CSV**
3. Select the dataset file
4. Click **Transform Data**
5. Power Query Editor opens automatically

*Screenshot Placeholder:*

(Insert screenshot of Power BI data import window)

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## 6. IDENTIFYING MISSING & BLANK VALUES

Using:

- Column Quality
- Column Distribution
- Column Profile

**Findings:**

| Column         | Issue Found  | Type    |
|----------------|--------------|---------|
| Views          | Blank values | Numeric |
| Likes          | Null values  | Numeric |
| Official_video | Null values  | Boolean |
| Description    | Blank text   | Text    |

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## 7. HANDLING MISSING, NULL & BLANK ENTRIES

### 7.1 Views Column (Blanks → 0)

Blank numeric fields cannot be used for:

- SUM
- AVG
- MAX
- Charts

Fix: Replace blank with **0**

Then convert to Whole Number.

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### 7.2 Likes Column (Null → 0)

Null likes break visualizations.

Fix: Replace null → 0.

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### 7.3 Official\_video Column

Contains TRUE/FALSE values.

Null was replaced with **0**.

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### 7.4 Description Column

Blank text allowed.

No changes needed.

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## 8. FIXING MERGED COLUMNS

Columns like:

- Spotify\_Info
- Youtube\_Info

contained multiple values inside one column using delimiters (|, --, ;).

### Steps to fix:

1. Select column
2. Split Column → By Delimiter
3. Choose delimiter
4. Trim and Clean
5. Remove prefixes
6. Rename split fields

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### ***Example Before:***

"Danceability--Energy--Tempo"

### ***Example After:***

- danceability
  - energy
  - tempo
-

## 9. STANDARDIZING COLUMN NAMES

Column names were inconsistent:

- Uppercase
- Lowercase
- Spaces
- Mixed cases

**Fix:**

Use consistent pattern:

- ✓ all lowercase
- ✓ use underscores
- ✓ avoid symbols

Examples:

- Track Name → track\_name
- ARTIST → artist
- Album-Type → album\_type

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## 10. TEXT CLEANING & FORMATTING

Applied text-cleaning functions:

| Function             | Purpose                   |
|----------------------|---------------------------|
| Trim                 | Removes extra spaces      |
| Clean                | Removes hidden characters |
| Capitalize Each Word | Standardizes names        |

Used on:

- artist
  - track
  - album
- 

## 11. REMOVING IRRELEVANT COLUMNS

Removed:

- Temporary split columns
  - System-generated index columns
  - Duplicate columns like Views\_orig
  - Random/noisy text columns
- 

## 12. CORRECTING INCONSISTENT DATA TYPES

Many numeric columns were stored as text:

- danceability
- energy
- acousticness
- valence
- tempo
- loudness
- duration\_ms

**Fix:**

- Convert to Decimal/Whole Number
  - Replace invalid text (N/A, --, unknown) with null → then 0
-

## 13. FIXING INVALID AND DIRTY ENTRIES

Invalid entries found:

### 13.1 In Views Column

- “invalid\_data”
- “error”  
→ replaced with 0

### 13.2 In Album Column

- Numeric album values  
→ replaced with “Unknown Album”
- 

## 14. REMOVING DUPLICATE ROWS

Method:

**Home → Remove Rows → Remove Duplicates**

Duplicates cause:

- Inflated view counts
- Wrong rankings

Final dataset contains **unique rows only**.

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## 15. REORDERING COLUMNS

Logical order applied:

**A. Track Metadata**

**B. YouTube Metrics**

**C. Spotify Audio Features**

**D. Additional Info**

Improves readability and modeling.

## 16. FINAL CLEAN DATASET SUMMARY

After cleaning:

- ✓ No nulls in numeric fields
  - ✓ All data types correct
  - ✓ All merged fields split
  - ✓ Invalid entries removed
  - ✓ Duplicates removed
  - ✓ Columns reordered
  - ✓ Dataset ready for dashboard creation
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## 17. CHALLENGES FACED

- Distinguishing blanks vs nulls
  - Removing invisible characters
  - Cleaning deeply merged columns
  - Converting text to numeric
  - Ensuring naming consistency
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## 18. CONCLUSION

This project demonstrates complete data cleaning using Power BI.  
The cleaned dataset is fully prepared for analysis, visualization, and building interactive dashboards.

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## 19. REFERENCES

- Microsoft Power BI Documentation
- Power Query Editor Transformation Guide